



Soil Respiration Calculations

CO₂ from Flux Chamber Measurements!



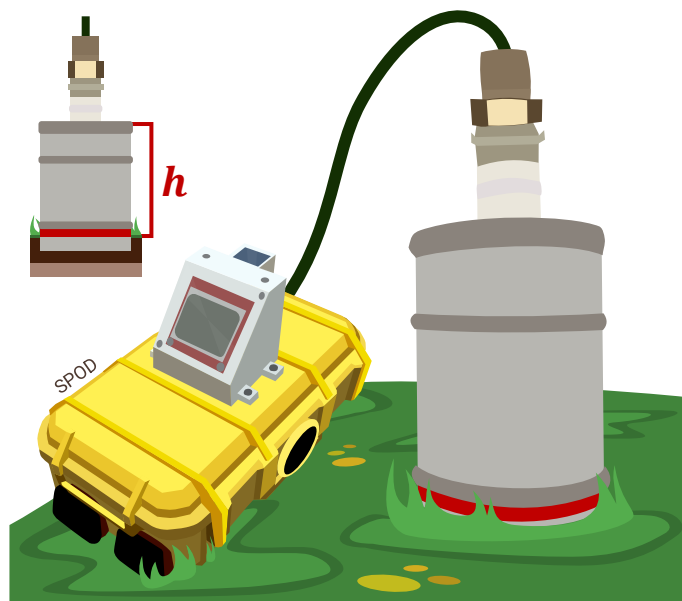
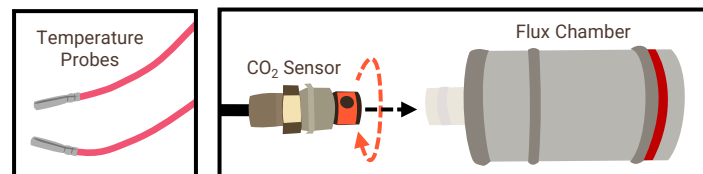
Experimental Setup Instructions:

1. Screw the CO₂ sensor into the flux chamber
2. Plug the CO₂ cable into the SPOD
3. Plug temperature probes into the SPOD
4. Turn on the SPOD & wait for the display to begin showing data
5. Push the flux chamber into the soil to the bottom of the red tape if possible
6. Put the temperature probes in the soil around the chamber

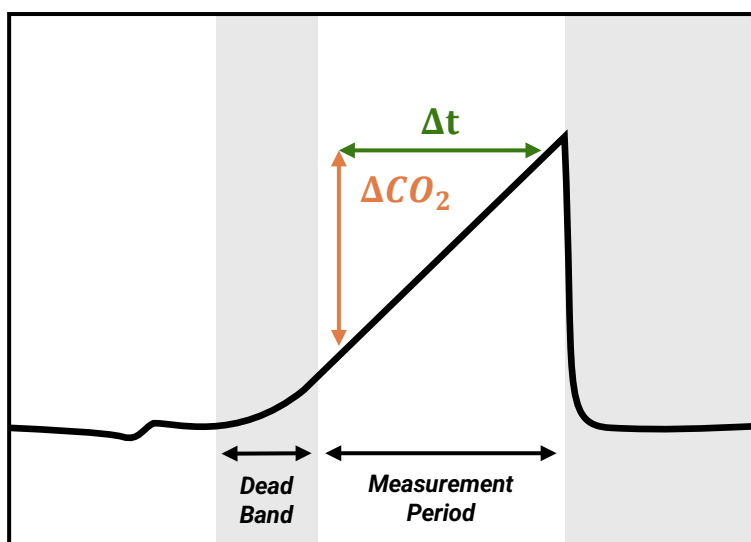
Measurement & Calculation Instructions:

Your measurements should follow the shape of the curve in the figure, refer to this for labels

1. Do not calculate with the Dead Band, you should wait approximately 5-10 seconds before taking measurements
2. Begin the Measurement Period, and allow the chamber to stay in the soil for 3-5 minutes
3. Find a linear segment in the Measurement Period & record ΔCO_2 & Δt



Variable	Value	Units
P_{atmos}	76,500	Pa
R	8.314	$\frac{m^3 \cdot Pa}{K \cdot mol}$
M_{CO_2}	44	$\frac{g}{mol} CO_2$
h	15.75	cm
T_{avg}		K
ΔCO_2		ppm
Δt		s
Φ_{CO_2}		$\frac{kg CO_2}{m^2 \cdot s}$



$$\Phi_{CO_2} = \frac{\Delta CO_2}{\Delta t} M h \frac{P_{atmos}}{R T_{avg}}$$

Based on Key Resources from:

Smith, V. (2026). Flux 101: A Beginner's Guide to Soil Greenhouse Gas Fluxes and Chamber Methods (Resources, Learn). Eosense Inc. <https://eosense.com/learn/soil-gas-flux-101.html>

Subke, J.-A., Kutzbach, L., & Risk, D. (2021). Soil Chamber Measurements. In T. Foken (Ed.), *Springer Handbook of Atmospheric Measurements* (pp. 1603–1624). Springer International Publishing.

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